

The Periodic Constantin–Lax–Majda Equation: Exact Arbitrary-Mean Flow and Complete First-Pole Clock

HCS-C377 theorem package

4 September 2026

Abstract

We solve the inviscid nonadvective Constantin–Lax–Majda equation on the circle for arbitrary conserved mean under one frozen periodic Hilbert transform convention. The periodic Tricomi identity turns the nonlocal PDE into a pointwise complex Riccati equation, while an invariant Hardy subspace proves that the Hilbert constraint is preserved. This yields exact Möbius formulas in both the zero- and nonzero-mean strata. For nonzero mean, forward smooth breakdown occurs if and only if the initial vorticity has a zero; the first time is an explicit compact minimum on the principal arccotangent branch. Zero-free data are global and periodic. A separate signed-zero criterion treats zero mean, and the full one-mode threshold diagram follows.

Keywords: periodic Hilbert transform; Constantin–Lax–Majda equation; complex Riccati flow; exact Möbius solution; first-pole clock; one-mode threshold.

中文摘要

本文在固定的周期希尔伯特变换符号约定下，求解任意守恒均值的无粘、无平流康斯坦丁、拉克斯和马吉达方程。周期特里科米恒等式把非局部方程化为逐点复黎卡提方程，而不变哈代子空间保证希尔伯特约束始终保持，由此得到零均值与非零均值两类精确莫比乌斯公式。当均值非零时，正向光滑解失效当且仅当初始涡量存在零点；首个失效时间由主值反余切分支上的紧致最小值精确给出。无零点数据全局存在并周期返回；零均值情形具有不同的带符号零点判据，单傅里叶模的全部阈值情形也被分类。

关键词：周期希尔伯特变换；康斯坦丁、拉克斯和马吉达方程；复黎卡提流；精确莫比乌斯解；首极点时钟；单模阈值。

1 Convention and exact Riccati closure

Write $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$. For a real smooth periodic function, define

$$\mathcal{H}(e^{ikx}) = -i \operatorname{sgn}(k) e^{ikx}, \quad \mathcal{H}1 = 0. \quad (1)$$

Thus $\mathcal{H} \sin(kx) = -\cos(kx)$ and $\mathcal{H} \cos(kx) = \sin(kx)$ for $k > 0$. On mean-zero functions, $\mathcal{H}^2 = -I$; on real L^2 , $\mathcal{H}$ is skew-adjoint.

Consider

$$\partial_t \omega = \omega \mathcal{H} \omega, \quad \omega(0, x) = \omega_0(x). \quad (2)$$

Let

$$\mu = \langle \omega_0 \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} \omega_0(x) dx, \quad f = \omega - \mu, \quad h = \mathcal{H}f, \quad z = h + if.$$

Lemma 1 (Periodic Tricomi identity with the fixed sign). *For real mean-zero f ,*

$$\mathcal{H}(f \mathcal{H}f) = \frac{(\mathcal{H}f)^2 - f^2}{2}. \quad (3)$$

Moreover $\langle f \mathcal{H}f \rangle = 0$.

Proof. For a trigonometric polynomial, split its Fourier series into positive and negative Hardy parts. On those parts (1) is multiplication by $-i$ and $+i$, respectively. Comparing the positive and negative parts of the product gives (3); the zero Fourier coefficient vanishes by the skew-adjointness of $\mathcal{H}$. Density and boundedness of $\mathcal{H}$ on Sobolev spaces extend the identity to smooth f . $\square$

Theorem 1 (Arbitrary-mean pointwise flow). *The mean μ is conserved and*

$$z_t = i\mu z + \frac{1}{2}z^2. \quad (4)$$

Before the first zero of its denominator, the exact solution is

$$\mu = 0 : \quad z(t, x) = \frac{2z_0(x)}{2 - tz_0(x)}, \quad (5)$$

$$\omega(t, x) = \frac{4\omega_0(x)}{[2 - th_0(x)]^2 + t^2\omega_0(x)^2}; \quad (6)$$

$$\mu \neq 0 : \quad z(t, x) = \frac{e^{i\mu t} z_0(x)}{1 - [e^{i\mu t} - 1]z_0(x)/(2i\mu)}. \quad (7)$$

If

$$\Delta(t, x) = 2i\mu - [e^{i\mu t} - 1]z_0(x), \quad (8)$$

then the real vorticity formula for $\mu \neq 0$ is

$$\omega(t, x) = \frac{4\mu^2\omega_0(x)}{|\Delta(t, x)|^2}. \quad (9)$$

Proof. Skew-adjointness gives $\partial_t \langle \omega \rangle = \langle \omega \mathcal{H} \omega \rangle = 0$. Hence

$$f_t = \mu h + fh, \quad h_t = \mu \mathcal{H} h + \mathcal{H}(fh) = -\mu f + \frac{1}{2}(h^2 - f^2),$$

where Lemma 1 and $\mathcal{H}h = -f$ were used. Combining the two real equations proves (4). Separation gives (5). For $\mu \neq 0$, differentiating $1/z$ gives a scalar linear equation; solving it yields (7). Taking the imaginary part of (5) gives (6). Finally multiply (7) by $2i\mu/\Delta$ and its complex conjugate. Using $z_0 = h_0 + i(\omega_0 - \mu)$ gives exactly (9).

It remains to check that the pointwise Riccati solution is still a solution of the nonlocal PDE, rather than merely of a larger product system. For $s > 1/2$, the negative-frequency Hardy subspace of $H^s(\mathbb{T})$ is a Banach algebra. Under (1), $z_0 = \mathcal{H}f_0 + if_0$ has only strictly negative Fourier modes. The vector field $z \mapsto i\mu z + z^2/2$ is locally Lipschitz and preserves that subspace, so uniqueness for the Banach-space ODE shows that the explicit $z(t)$ retains strictly negative modes. Consequently its imaginary part has mean zero and its real part is exactly the Hilbert transform of its imaginary part: $h = \mathcal{H}f$.

Equivalently, if $g = h - \mathcal{H}f$ is the constraint defect in the product system, Lemma 1 gives

$$g_t = (\mathcal{H}f)g + \frac{1}{2}g^2 - \mu \mathcal{H}g - \mathcal{H}(fg).$$

This equation is locally Lipschitz in the same Sobolev algebra, and $g(0) = 0$ forces $g \equiv 0$. On any compact time interval where the displayed denominator has a positive lower bound, the explicit rational formula remains smooth in x ; the same local uniqueness argument continues the PDE solution across the interval. Thus a denominator zero, not a loss of the Hilbert constraint, is the only obstruction used below. $\square$

Round-zero certificate: round zero fixes the periodic Hilbert sign, proves the Tricomi closure, conserves the arbitrary mean, and derives both exact Möbius solutions and real vorticity formulas.

2 Complete forward-time classification

Use the principal branch

$$\operatorname{arccot} : \mathbb{R} \longrightarrow (0, \pi), \quad \frac{d}{dr} \operatorname{arccot} r = -\frac{1}{1 + r^2}. \quad (10)$$

Theorem 2 (Nonzero mean: zero set if and only if breakdown). *Suppose $\mu \neq 0$, put $a = |\mu|$, and let $Z_0 = \{x : \omega_0(x) = 0\}$. If Z_0 is empty, the solution is global, smooth, and has time period $2\pi/a$ (not necessarily its least period). If Z_0 is nonempty, the first positive denominator pole is*

$$T_+ = \min_{x \in Z_0} \frac{2}{a} \operatorname{arccot}(h_0(x)/a), \quad (11)$$

and the classical smooth solution breaks down at T_+ . Thus forward breakdown occurs if and only if the initial vorticity has a zero.

Proof. For $0 < t < 2\pi/a$, the equation $\Delta(t, x) = 0$ is equivalent to

$$z_0(x) = \frac{2i\mu}{e^{i\mu t} - 1} = \mu \cot(\mu t/2) - i\mu.$$

Its imaginary part is $\omega_0(x) - \mu = -\mu$, exactly the condition $\omega_0(x) = 0$. Its real part is

$$h_0(x) = \mu \cot(\mu t/2) = a \cot(at/2).$$

The last function decreases bijectively from $+\infty$ to $-\infty$ on the displayed time interval. Equation (10) therefore gives one and only one pole time for each initial zero, and compactness of Z_0 gives the minimum (11). At a pole, z has a Riccati pole, so its real part $h = \mathcal{H}(\omega - \mu)$ becomes unbounded and smooth evolution fails. If Z_0 is empty, Δ never vanishes. It is periodic in time and bounded away from zero on one compact time cell, proving global smoothness and the stated period. $\square$

Theorem 3 (Zero mean: signed-zero clock). *Suppose $\mu = 0$ and define*

$$Z_+ = \{x : \omega_0(x) = 0, h_0(x) > 0\}.$$

If Z_+ is nonempty, the first positive denominator pole is

$$T_+ = \min_{x \in Z_+} \frac{2}{h_0(x)}. \quad (12)$$

If Z_+ is empty, the exact solution is global forward in time. In particular, the identically zero datum is stationary.

Proof. The denominator in (6) is a sum of two real squares. It vanishes precisely when $\omega_0(x) = 0$ and $2 - th_0(x) = 0$. A positive solution requires $h_0(x) > 0$ and is then unique. Taking the first such time proves (12); if there is none, no positive-time denominator can vanish. $\square$

3 Complete one-mode phase diagram

Corollary 1 (One Fourier mode). *Let $A \neq 0, k \in \mathbb{Z}_{>0}$, and*

$$\omega_0(x) = \mu + A \sin(kx), \quad h_0(x) = -A \cos(kx). \quad (13)$$

For $\mu \neq 0$:

$$\begin{aligned} |\mu| > |A| : & \text{ global and periodic;} \\ |\mu| = |A| : & T_+ = \pi/|\mu| \quad (\text{tangent zero}); \\ 0 < |\mu| < |A| : & T_+ = \frac{2}{|\mu|} \operatorname{arccot}(\sqrt{A^2 - \mu^2}/|\mu|). \end{aligned}$$

For $\mu = 0, T_+ = 2/|A|$. The clock is independent of k ; the number of spatial copies changes with k .

Proof. Zeros exist exactly when $|\mu| \leq |A|$. At a zero, $\sin(kx) = -\mu/A$, so the two possible Hilbert values are $h_0 = \pm\sqrt{A^2 - \mu^2}$. Since (10) is decreasing, the positive value gives the first time in the crossing regime. At equality the Hilbert value is zero. For zero mean, the largest positive Hilbert value at a zero is $|A|$, and Theorem 3 applies. $\square$

Round-one certificate: round one adds necessary-and- sufficient global/breakdown criteria for both mean strata, fixes the principal arccot branch, proves the first forward clock, and closes the entire one-mode phase diagram.